

CAM LAB REPORT

by

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ME (Cloud Computing)**



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(A Constituent unit of MAHE, Manipal)

Lab Experiment 1

Configuring Passwordless SSH Authentication

1. Master and slave nodes are identified and IPs are named accordingly in the file /etc/hosts.

172.18.181.65 master

172.18.181.61 slave1

172.18.181.53 slave2

2. Generating RSA key

ssh-keygen -t rsa -b 4096

```
msis@msis:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/msis/.ssh/id_rsa):
/home/msis/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/msis/.ssh/id_rsa
Your public key has been saved in /home/msis/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:dDk1kj+unxgcvCrSatcQ5Vrnypmwiv+XHR6Mea8tS4g msis@msis.kubenode1.com
The key's randomart image is:
+---[RSA 4096]---+
|                |
|      ..o       |
|      . .+ .    |
|      o. +.     |
|      ..oo..o   |
|      +S*o. .   |
|      +.oo*o.   |
|      .E*.0==   |
|      .o = 0o=+..|
|      .o+*.o. +=+|
+---[SHA256]-----+
msis@msis:~$
```

3. Once the key is generated, copy or push it to slave nodes.

ssh-copy-id msis@172.18.181.61

```
msis@msis:~$ ssh-copy-id msis@172.18.181.61
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
msis@172.18.181.61's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'msis@172.18.181.61'"
and check to make sure that only the key(s) you wanted were added.

msis@msis:~$
```

ssh-copy-id user@172.18.181.53

```
msis@msis:~$ ssh-copy-id msis@172.18.181.53
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
msis@172.18.181.53's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'msis@172.18.181.53'"
and check to make sure that only the key(s) you wanted were added.

msis@msis:~$
```

The above commands copies your public key (~/.ssh/id_rsa.pub) to the slave's ~/.ssh/authorized_keys.

Making sure that the SSH agent is running and the key is loaded eval "\$(ssh-agent -s)" --> Starts the agent
ssh-add ~/.ssh/id_rsa --> Adding private key to the agent

Giving the necessary permissions chmod 600 ~/.ssh/id_rsa --> Private RSA key (Only accessible by owner)
chmod 644 ~/.ssh/id_rsa.pub --> Public RSA key (Shared to slaves)
~/.ssh/authorized_keys)

4. Verifying the passwordless ssh connection

ssh msis@172.18.181.61

```
msis@msis:~$ ssh msis@172.18.181.61
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.14.0-27-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

197 updates can be applied immediately.
104 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Tue Nov 11 21:47:10 2025 from 172.18.181.70
msis@msis:~$
```

ssh [msis@172.18.181.53](#)

```
msis@msis:~$ ssh msis@172.18.181.53
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.14.0-27-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

186 updates can be applied immediately.
95 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Sat Nov 15 14:34:26 2025 from 172.18.181.61
msis@msis:~$
```

Lab Experiment 2

Configuration & Implementation of NFS Server–Client File Sharing in Linux

1. Install NFS Packages on All Nodes

On master node

- `sudo apt update`
- `sudo apt install nfs-kernel-server -y`

On slave nodes

- `sudo apt update`
- `sudo apt install nfs-common -y`

2. Create a Shared Directory on the Master

- `sudo mkdir -p /nfs/shared`
- `sudo chmod 777 /nfs/shared`

```
msis@msis:~$ sudo mkdir -p /nfs/shared
[sudo] password for msis:
msis@msis:~$ sudo chmod 777 /nfs/shared
msis@msis:~$ cd /nfs
msis@msis:/nfs$ ls
shared
msis@msis:/nfs$
```

3. Configure NFS Exports on the Master

sudo nano /etc/exports

Add the following lines (slave IPs):

- /nfs/shared 172.18.181.61(rw,sync,no_subtree_check)
- /nfs/shared 172.18.181.53(rw,sync,no_subtree_check)

```
GNU nano 7.2 /etc/exports *
# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes      hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4       gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes gss/krb5i(rw,sync,no_subtree_check)
#
/home/msis/storage *(rw,sync,no_root_squash,no_subtree_check)
/nfs/shared 172.18.181.61(rw,sync,no_subtree_check)
/nfs/shared 172.18.181.53(rw,sync,no_subtree_check)

```

4. Export the Shared Directory and Apply Configuration

- sudo exportfs -a
- sudo systemctl restart nfs-kernel-server
- sudo exportfs -v

```
msis@msis:/nfs/shared$ sudo exportfs -a
msis@msis:/nfs/shared$ sudo systemctl restart nfs-kernel-server
msis@msis:/nfs/shared$ sudo exportfs -v
/nfs/shared      172.18.181.61(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)
/nfs/shared      172.18.181.53(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)
/home/msis/storage
                  <world>(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,no_root_squash,no_all_squash)
msis@msis:/nfs/shared$

```

5. Mount the NFS Share on Slave Nodes

On Slave 1

- `sudo mkdir -p /nfs/shared`
- `sudo mount 172.18.181.65:/nfs/shared /nfs/shared`

```
msis@msis:~$ ssh msi@172.18.181.61
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.14.0-27-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

197 updates can be applied immediately.
104 of these updates are standard security updates.
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1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Sat Nov 15 15:01:36 2025 from 172.18.181.53
msis@msis:~$ sudo apt install nfs-common -y
[sudo] password for msi:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nfs-common is already the newest version (1:2.6.4-3ubuntu5.1).
0 upgraded, 0 newly installed, 0 to remove and 213 not upgraded.
msis@msis:~$ cd /nfs
msis@msis:/nfs$ ls
shared
msis@msis:/nfs$ sudo rm -rf shared
msis@msis:/nfs$ ls
msis@msis:/nfs$ sudo mkdir -p /nfs/shared
msis@msis:/nfs$ sudo mount 172.18.181.65:/nfs/shared /nfs/shared
msis@msis:/nfs$
```

On Slave 2

- `sudo mkdir -p /nfs/shared`
- `sudo mount 172.18.181.65:/nfs/shared /nfs/shared`

```
msis@msis:~$ ssh msi@172.18.181.53
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.14.0-27-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

186 updates can be applied immediately.
95 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Sat Nov 15 15:04:04 2025 from 172.18.181.53
msis@msis:~$ sudo apt install nfs-common -y
[sudo] password for msi:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nfs-common is already the newest version (1:2.6.4-3ubuntu5.1).
0 upgraded, 0 newly installed, 0 to remove and 199 not upgraded.
msis@msis:~$ cd /nfs
msis@msis:/nfs$ ls
shared
msis@msis:/nfs$ sudo rm -rf shared/
msis@msis:/nfs$ sudo mkdir -p /nfs/shared
msis@msis:/nfs$ sudo mount 172.18.181.65:/nfs/shared /nfs/shared
msis@msis:/nfs$
```


6. Verify the NFS Mount

`df -h | grep nfs`

```
msis@msis:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                     1.6G      2.3M    1.6G   1% /run
/dev/nvme0n1p2            937G     325G     566G  37% /
tmpfs                     7.7G         0    7.7G   0% /dev/shm
tmpfs                     5.0M      8.0K    5.0M   1% /run/lock
efivarfs                 246K     155K     87K  65% /sys/firmware/efi/efivars
/dev/nvme0n1p1            1.1G      6.2M    1.1G   1% /boot/efi
tmpfs                     1.6G     112K    1.6G   1% /run/user/1000
172.18.181.65:/nfs/shared 937G     159G     731G  18% /nfs/shared
msis@msis:~$
```

7. Test File Sharing Between Nodes

On any slave machine

- `cd /nfs/shared`
- `touch testfile_from_slave1`

On slave

```
msis@msis:~$ cd /nfs/shared
msis@msis:/nfs/shared$ touch nfs_file_from_slave1
msis@msis:/nfs/shared$ ls
nfs_file_from_slave1  nfsssss  nfstesting
msis@msis:/nfs/shared$
```

On Master

```
msis@msis:~$ cd /nfs/shared
msis@msis:/nfs/shared$ ls
nfs_file_from_slave1  nfsssss  nfstesting
msis@msis:/nfs/shared$
```

Lab Experiment 3

TrueNAS NFS, SMB and WebDAV Implementation Lab

NFS Setup in TrueNAS

1. Create a Dataset

- Go to Storage → Pools
- Click : (3 dots) next to the pool → Add Dataset
- Give a name → e.g., nfspool
- Set:
 - Share Type: Generic
 - Case Sensitivity: Sensitive
- Save

2. Set Permissions

- Go to Storage → Datasets
- Select nfs_share
- Click Edit Permissions
- Choose:
 - User: root
 - Group: wheel
 - Mode: Read/Write
- Apply

3. Configure NFS Service

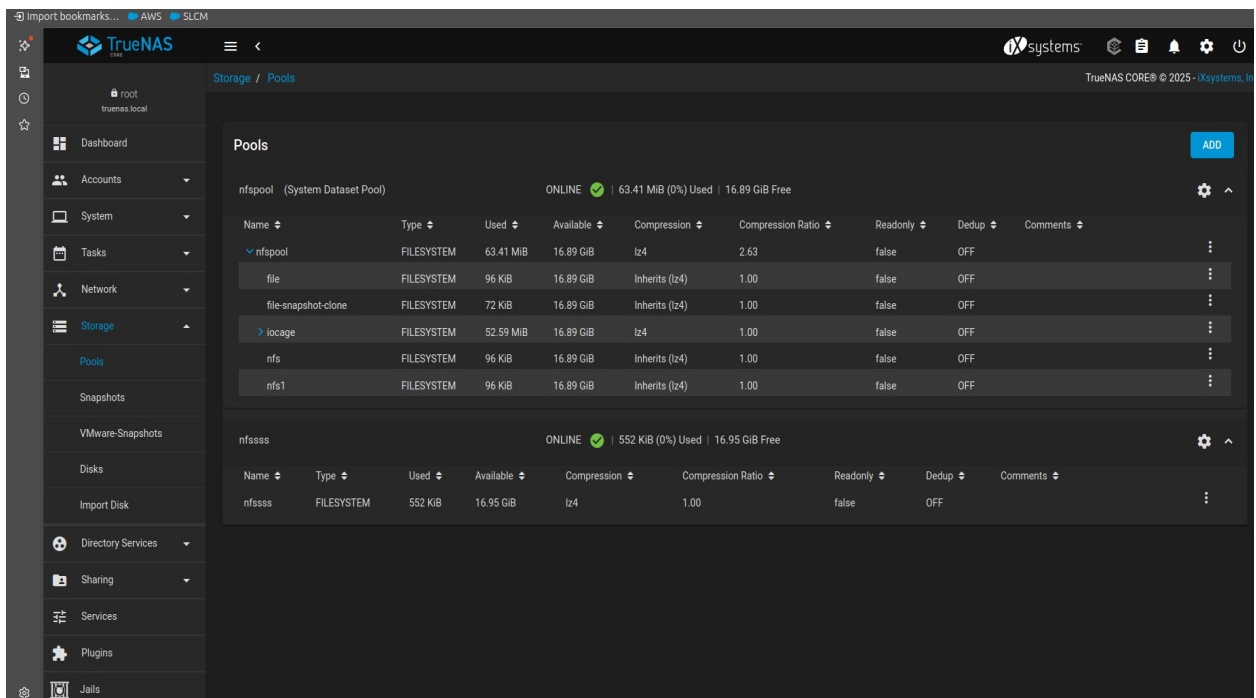
- Go to Services
- Enable NFS
- Click the wrench icon:
 - Check Enable NFSv4
 - Number of servers: default
 - Bind IPs
- Save and start the service

4. Create NFS Share

- Go to Shares → Unix Shares (NFS)
- Click Add
- Set:
 - Path: /mnt/nfspool/nfs_share
 - Maproot User: root or nobody
 - Maproot Group: nogroup
 - Allowed networks or hosts: your client IPs
- Save and enable

5. Mount NFS on Client (Linux)

- `sudo apt install nfs-common -y`
- `sudo mkdir -p /mnt/nfs`
- `sudo mount -t nfs -o nfsvers=4 192.168.101.130:/mnt/nfspool /mnt/nfs`



Snapshot of name, file-snapshot-clone has been created and assigned to the user gagana.

WebDAV Setup in TrueNAS

1. Create Dataset

- Storage → Add Dataset
- Name it students
- Save

2. Set Permissions

- Select webdav
- Edit Permissions
- Select:
 - User: root
 - Group: wheel
 - Mode: Read / Write
- Save

3. Create WebDAV Share

- Shares → WebDAV → Add
- Path: /mnt/webpool/webdav
- Enable:
 - Read/Write
 - Authentication Required
- Save

4. Enable WebDAV Service

- Services → WebDAV
- Enable Service
- Configure:
 - Protocol: HTTP + HTTPS
 - Listen IP: All

Start the service

TrueNAS

systems

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root
truenas.local

Dashboard

Accounts

System

Tasks

Network

Storage

Pools

Snapshots

VMware-Snapshots

Disks

Import Disk

Directory Services

Sharing

Services

Plugins

Jails

Reporting

Storage / Pools

Pools

ADD

smbspool

ONLINE ✓ | 760 KiB (0%) Used | 16.95 GiB Free

Name	Type	Used	Available	Compression	Compression Ratio	Readonly	Dedup	Comments
smbspool	FILESYSTEM	760 KiB	16.95 GiB	lz4	1.00	false	OFF	
fil2	FILESYSTEM	96 KiB	16.95 GiB	Inherits (lz4)	1.00	false	OFF	

webpool (System Dataset Pool)

ONLINE ✓ | 9.75 MiB (0%) Used | 16.94 GiB Free

Name	Type	Used	Available	Compression	Compression Ratio	Readonly	Dedup	Comments
webpool	FILESYSTEM	9.75 MiB	16.94 GiB	lz4	16.43	false	OFF	
students	FILESYSTEM	464 KiB	16.94 GiB	Inherits (lz4)	1.00	false	OFF	

TrueNAS

systems

TrueNAS CORE © 2025 - iXsystems, Inc.

root
truenas.local

Dashboard

Accounts

System

Tasks

Network

Storage

Directory Services

Sharing

Apple Shares (AFP)

Block Shares (SCSI)

Unix Shares (NFS)

WebDAV Shares

Windows Shares (SMB)

Services

Plugins

Jails

Reporting

Sharing / WebDAV

WebDAV

Filter WebDAV

COLUMNS

ADD

Share Name	Description	Path	Enabled
webdav		/mnt/webpool	yes

1 - 1 of 1

Index of /webdav

- [Parent Directory](#)
- [file/](#)
- [kush/](#)
- [kushal/](#)
- [students/](#)

SMB (Windows File Sharing) Setup in TrueNAS

1. Create a Dataset

- Storage → Pools
- Add Dataset → name it smbpool
- Set:
 - Share Type: 5MB
- Save

2. Set Permissions

- Storage → Datasets → Select smbpool
- Edit Permissions
- Choose a user/group:
 - User: root
 - Group: wheel
- Save

3. Enable SMB Service

- Services → SMB → Enable
- Configure:
 - Workgroup: WORKGROUP
 - Enable SMB2/SMB3
- Save & start

4. Create SMB Share

- Shares → Windows Shares (SMB) → Add
- Path: /mnt/smbpool
- Name: smbshare
- Enable:
 - Allow Guest Access (optional)
 - Browsable
- Save

5. Access Share

Run in windows File Explorer

192.168.101.130\smbpool

TrueNAS

root
truenas.local

Dashboard

Accounts

System

Tasks

Network

Storage

Pools

Snapshots

VMware-Snapshots

Disks

Import Disk

Directory Services

Sharing

Services

Plugins

Jails

Reporting

Storage / Pools

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Pools

ADD

smbpool
ONLINE 760 KiB (0%) Used | 16.95 GiB Free

Name	Type	Used	Available	Compression	Compression Ratio	Readonly	Dedup	Comments
smbpool	FILESYSTEM	760 KiB	16.95 GiB	lz4	1.00	false	OFF	
fil2	FILESYSTEM	96 KiB	16.95 GiB	Inherits (lz4)	1.00	false	OFF	

webpool (System Dataset Pool)
ONLINE 9.75 MiB (0%) Used | 16.94 GiB Free

Name	Type	Used	Available	Compression	Compression Ratio	Readonly	Dedup	Comments
webpool	FILESYSTEM	9.75 MiB	16.94 GiB	lz4	16.43	false	OFF	
students	FILESYSTEM	464 KiB	16.94 GiB	Inherits (lz4)	1.00	false	OFF	

TrueNAS

root
truenas.local

Dashboard

Accounts

System

Tasks

Network

Storage

Directory Services

Sharing

Apple Shares (AFP)

Block Shares (iSCSI)

Unix Shares (NFS)

WebDAV Shares

Windows Shares (SMB)

Services

Plugins

Jails

Reporting

Sharing / SMB

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Samba

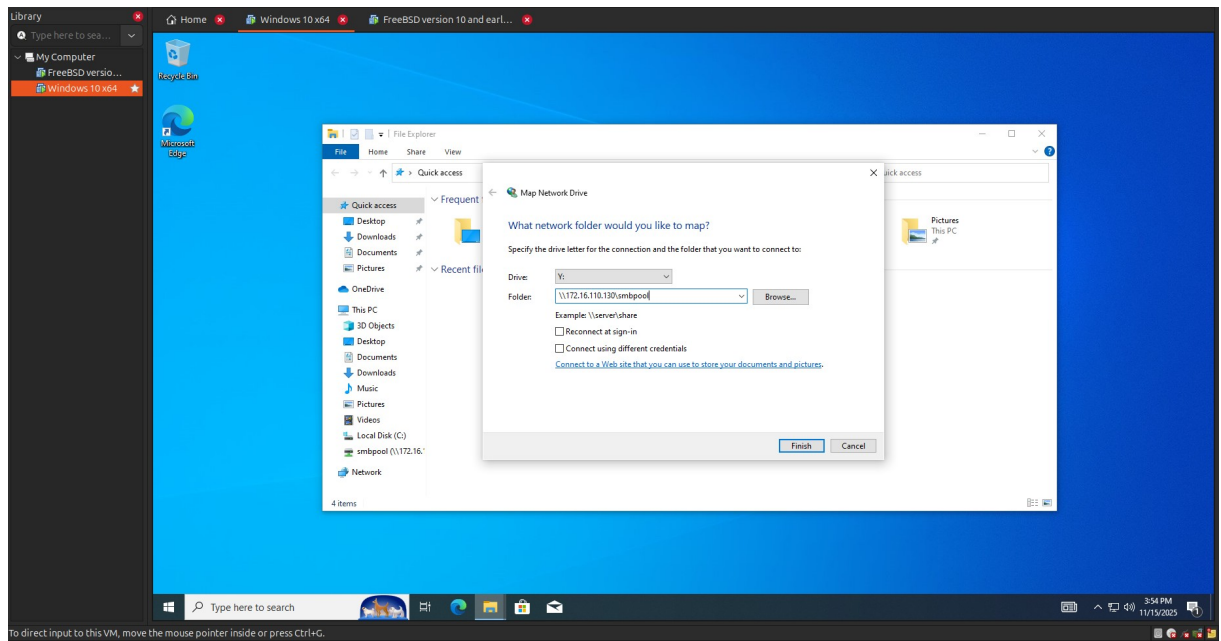
Filter Samba

COLUMNS

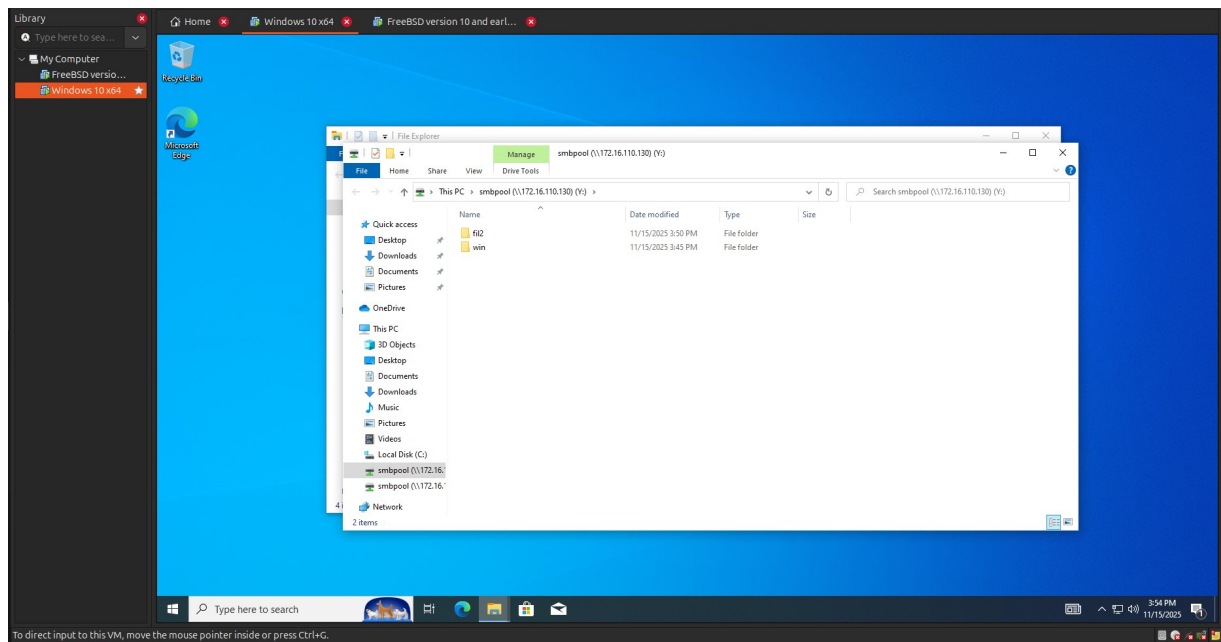
ADD

Name	Path	Description	Enabled
smbpool	/mnt/smbpool		yes

1 - 1 of 1



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Lab Experiment 4

Implementation of Web Services using Java JAX-WS, Tomcat, and IntelliJ IDEA

Aim

To develop, publish, and consume a SOAP-based web service using Java JAX-WS, Tomcat server, and IntelliJ IDEA, and test the functionality through a Java client application.

Software Requirements

- Ubuntu Linux
- JDK 8 (OpenJDK 8)
- IntelliJ IDEA Ultimate
- Apache Tomcat (installed under /opt/tomcat or /opt/tomcat9)
- Maven
- wsimport tool (comes with JDK 8)

Procedure

Part A: Setting Up the Development Environment

1. Install JDK 8 and IntelliJ IDEA Ultimate

- `sudo apt update`
- `sudo apt install openjdk-8-jdk -y`
- `sudo snap install intellij-idea-ultimate --classic`

2. Check Tomcat Installation

- `sudo systemctl status tomcat`
- `sudo chmod -R a+rX /opt/tomcat9`

Part B: Creating and Publishing the Web Service

1. Configure Tomcat in IntelliJ

In IntelliJ:

- File → Settings
- Build, Execution, Deployment → Application Servers
- Click +
- Select Tomcat Server
- Set Tomcat Home: /opt/tomcat

2. Create a New Web Application Project

- Go to File → New Project
- On the left select Java Enterprise or Jakarta EE
- Choose Web Application
- Project Name: calculator
- App Server: Tomcat
- Finish

3. Configure Project SDK & Bytecode Levels

Project Structure

- File → Project Structure → Modules
- Language Level: 8 - Lambdas, type annotations
- Dependencies → SDK 1.8

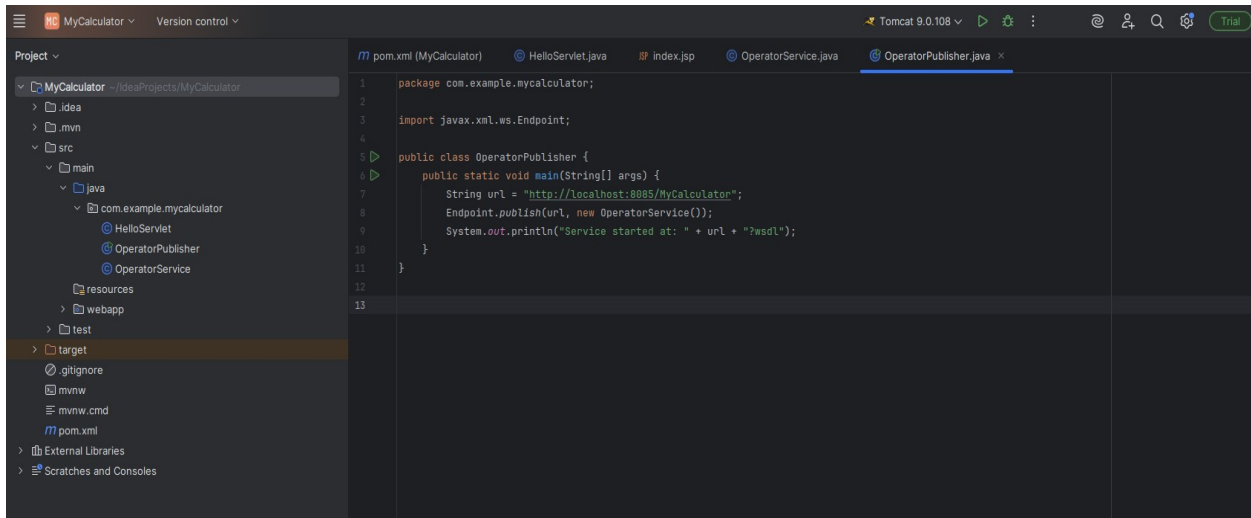
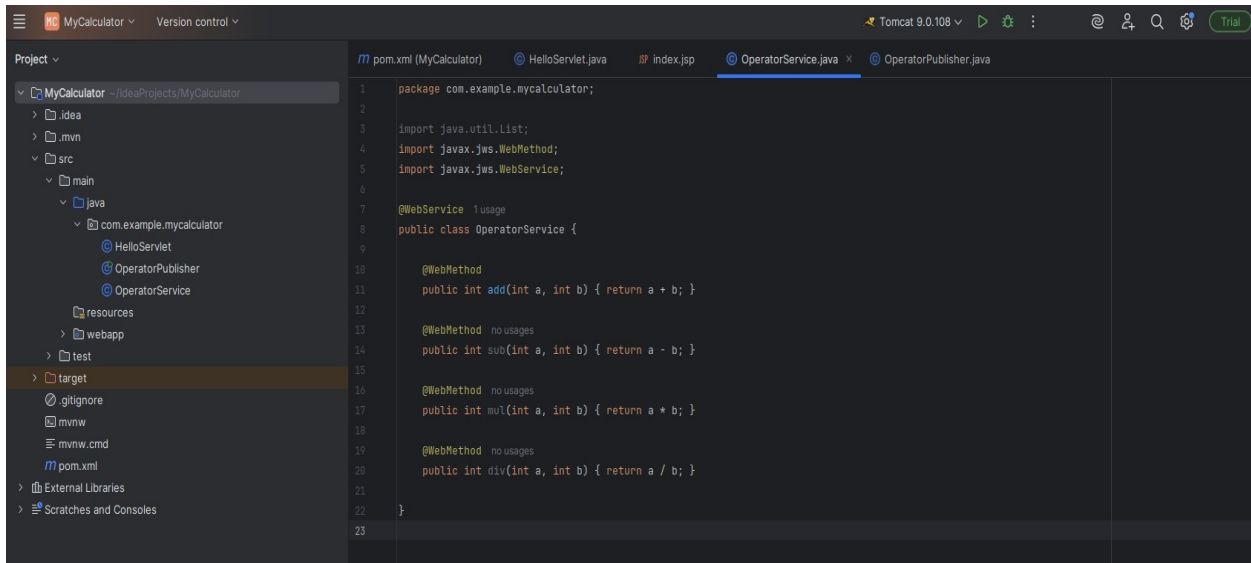
Compiler Settings

- File → Settings → Build → Compiler → Java Compiler
- Per-module bytecode version → set to 8
- Build Tools → Maven → Runner → JRE Java 1.8

4. Add Service Classes

Inside src/main/java/com.example.calculator

Create OperatorService.java class file and OperatorPublisher.java class file.



5. Configure pom.xml

Ensure these properties exist

```
<properties>
  <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  <maven.compiler.target>1.8</maven.compiler.target>
  <maven.compiler.source>1.8</maven.compiler.source>
  <junit.version>5.13.2</junit.version>
</properties>

<dependencies>
  <dependency>
    <groupId>jakarta.servlet</groupId>
    <artifactId>jakarta.servlet-api</artifactId>
    <version>4.0.4</version>
    <scope>provided</scope>
  </dependency>
</dependencies>
```

6. Build & Run the Service

Build → Rebuild Project → Run OperatorPublisher.java

Service started at: <http://localhost:8085/MyCalculator?wsdl> (Open in browser). This confirms that the SOAP service is running.

Part C: Creating the SOAP Client

1. Create a Standalone Java Client Project

- File → New Project
- Choose Java (NOT Java Enterprise)
- Name it client (open in new window)

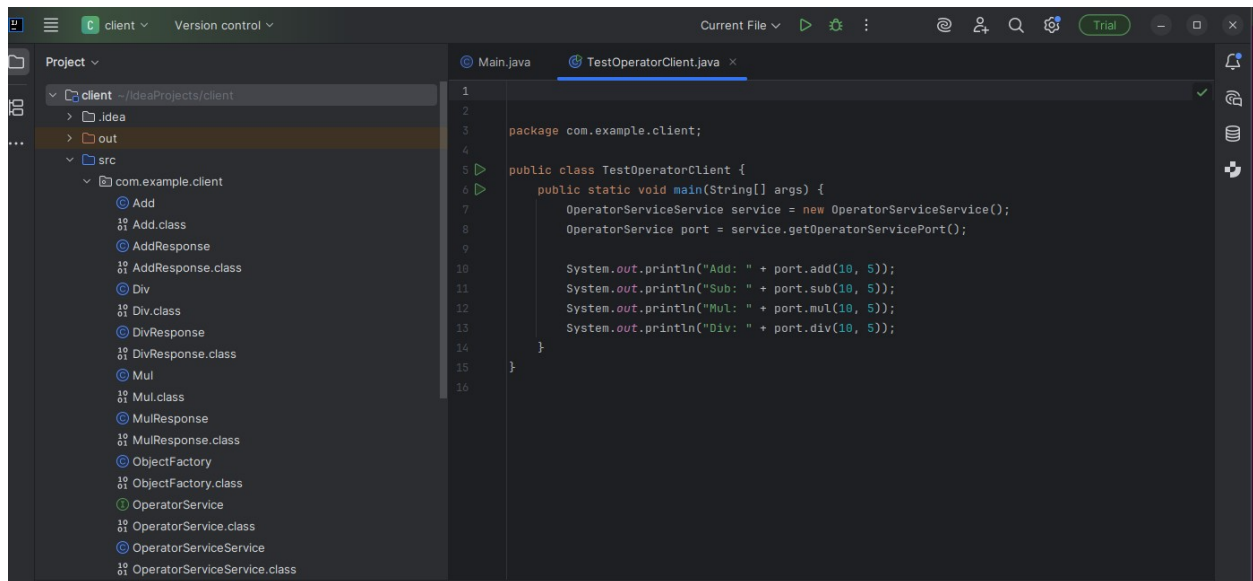
2. Generate Client Stubs Using wsimport

In the terminal

- `cd ~/IdeaProjects/client/src/`
- `wsimport -keep -p com.example.client http://localhost:8085/calculator?wsdl`

3. Create TestOperatorClient.java

Inside src/com/example/client, create TestOperatorClient.java file



Output:

